MSG No. 179.166 P_c6_522 [Type IV, hexagonal]

Table 1: Wyckoff site: 6a, site symmetry: 2'2'2

No.	position	mapping
1	$\left[0,\ 0,\ \frac{3}{4}\right]$	[1,11,16,19]
2	$\left[0,\ 0,\ \frac{7}{12}\right]$	[2,9,17,22]
3	$\left[0,\ 0,\ \frac{5}{12}\right]$	[3,12,18,20]
4	$\left[0,\ 0,\ \frac{1}{4}\right]$	[4,7,13,23]
5	$\left[0,\ 0,\ \frac{1}{12}\right]$	[5,10,14,21]
6	$\left[0,\ 0,\ \frac{11}{12}\right]$	[6,8,15,24]

Table 2: Wyckoff site: 6b, site symmetry: 2'22'

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,7,16,23]
2	$\left[0,\ 0,\ \frac{5}{6}\right]$	[2,10,17,21]
3	$\left[0,\ 0,\ \frac{2}{3}\right]$	[3,8,18,24]
4	$\left[0,\ 0,\ \frac{1}{2}\right]$	[4,11,13,19]
5	$\left[0,\ 0,\ \frac{1}{3}\right]$	[5,9,14,22]
6	$\left[0,\ 0,\ \frac{1}{6}\right]$	[6,12,15,20]

Table 3: Wyckoff site: 6c, site symmetry: 2'2'2

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[1,11,16,19]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{7}{12}\right]$	[2,9,17,22]
3	$\left[0,\ \frac{1}{2},\ \frac{5}{12}\right]$	[3,12,18,20]
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[4,7,13,23]
5	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{12}\right]$	[5,10,14,21]
6	$\left[0,\ \frac{1}{2},\ \frac{11}{12}\right]$	[6,8,15,24]

Table 4: Wyckoff site: 6d, site symmetry: 2'22'

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	[1,7,16,23]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{5}{6}\right]$	[2,10,17,21]
3	$\left[0,\ \frac{1}{2},\ \frac{2}{3}\right]$	[3,8,18,24]
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[4,11,13,19]
5	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{3}\right]$	[5,9,14,22]
6	$\left[0,\ \frac{1}{2},\ \frac{1}{6}\right]$	[6,12,15,20]

Table 5: Wyckoff site: `12e`, site symmetry: `2'..`

No.	position	mapping
1	$\left[0,\ 0,\ z\right]$	`[1,16]`
2	$\left[0,\ 0,\ z+\frac{5}{6}\right]$	`[2,17]`
3	$\left[0,\ 0,\ z+\frac{2}{3}\right]$	`[3,18]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4,13]`
5	$\left[0,\ 0,\ z+\frac{1}{3}\right]$	`[5,14]`
6	$\left[0,\ 0,\ z+\frac{1}{6}\right]$	`[6,15]`
7	$\left[0,\ 0,\ -z\right]$	`[7,23]`
8	$\left[0,\ 0,\ \frac{2}{3}-z\right]$	`[8,24]`
9	$\left[0,\ 0,\ \frac{1}{3}-z\right]$	`[9,22]`
10	$\left[0,\ 0,\ \frac{5}{6}-z\right]$	`[10,21]`
11	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[11,19]`
12	$\left[0,\ 0,\ \frac{1}{6}-z\right]$	`[12,20]`

Table 6: Wyckoff site: `12f`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{5}{6}\right]$	`[2,17]`
3	$\left[0,\ \frac{1}{2},\ z+\frac{2}{3}\right]$	`[3,18]`
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[4,13]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{3}\right]$	`[5,14]`
6	$\left[0,\ \frac{1}{2},\ z+\frac{1}{6}\right]$	`[6,15]`
7	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,23]`
8	$\left[0,\ \frac{1}{2},\ \frac{2}{3}-z\right]$	`[8,24]`
9	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{3}-z\right]$	`[9,22]`
10	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{5}{6}-z\right]$	`[10,21]`
11	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[11,19]`
12	$\left[0,\ \frac{1}{2},\ \frac{1}{6}-z\right]$	`[12,20]`

Table 7: Wyckoff site: `12g`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[1,19]`
2	$\left[x,\ x,\ \frac{7}{12}\right]$	`[2,22]`
3	$\left[0,\ x,\ \frac{5}{12}\right]$	`[3,20]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,23]`
5	$\left[-x,\ -x,\ \frac{1}{12}\right]$	`[5,21]`
6	$\left[0,\ -x,\ \frac{11}{12}\right]$	`[6,24]`
7	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[7,13]`
8	$\left[0,\ x,\ \frac{11}{12}\right]$	`[8,15]`
9	$\left[-x,\ -x,\ \frac{7}{12}\right]$	`[9,17]`
10	$\left[x,\ x,\ \frac{1}{12}\right]$	`[10,14]`
11	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[11,16]`
12	$\left[0,\ -x,\ \frac{5}{12}\right]$	`[12,18]`

Table 8: Wyckoff site: 12h, site symmetry: .2.

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,7]
2	$\left[x,\ x,\ \frac{5}{6}\right]$	[2,10]
3	$\left[0,\ x,\ \frac{2}{3}\right]$	[3,8]
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[4,11]
5	$\left[-x,\ -x,\ \frac{1}{3}\right]$	[5,9]
6	$\left[0,\ -x,\ \frac{1}{6}\right]$	[6,12]
7	$\left[x,\ 0,\ \frac{1}{2}\right]$	[13,19]
8	$\left[x,\ x,\ \frac{1}{3}\right]$	[14,22]
9	$\left[0,\ x,\ \frac{1}{6}\right]$	[15,20]
10	$[-x,\ 0,\ 0]$	[16,23]
11	$\left[-x,\ -x,\ \frac{5}{6}\right]$	[17,21]
12	$\left[0,\ -x,\ \frac{2}{3}\right]$	[18,24]

Table 9: Wyckoff site: 12i, site symmetry: ..2

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{3}{4}\right]$	[1,11]
2	$\left[-x,\ x,\ \frac{7}{12}\right]$	[2,9]
3	$\left[-2x,\ -x,\ \frac{5}{12}\right]$	[3,12]
4	$\left[-x,\ -2x,\ \frac{1}{4}\right]$	[4,7]
5	$\left[x,\ -x,\ \frac{1}{12}\right]$	[5,10]
6	$\left[2x,\ x,\ \frac{11}{12}\right]$	[6,8]
7	$\left[x,\ 2x,\ \frac{1}{4}\right]$	[13,23]
8	$\left[-x,\ x,\ \frac{1}{12}\right]$	[14,21]
9	$\left[-2x,\ -x,\ \frac{11}{12}\right]$	[15,24]
10	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	[16,19]
11	$\left[x,\ -x,\ \frac{7}{12}\right]$	[17,22]
12	$\left[2x,\ x,\ \frac{5}{12}\right]$	[18,20]

Table 10: Wyckoff site: 12j, site symmetry: ..2'

No.	position	mapping
1	$[x,\ 2x,\ 0]$	[1,23]
2	$\left[-x,\ x,\ \frac{5}{6}\right]$	[2,21]
3	$\left[-2x,\ -x,\ \frac{2}{3}\right]$	[3,24]
4	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	[4,19]
5	$\left[x,\ -x,\ \frac{1}{3}\right]$	[5,22]
6	$\left[2x,\ x,\ \frac{1}{6}\right]$	[6,20]
7	$[-x,\ -2x,\ 0]$	[7,16]
8	$\left[2x,\ x,\ \frac{2}{3}\right]$	[8,18]
9	$\left[-x,\ x,\ \frac{1}{3}\right]$	[9,14]
10	$\left[x,\ -x,\ \frac{5}{6}\right]$	[10,17]
11	$\left[x,\ 2x,\ \frac{1}{2}\right]$	[11,13]
12	$\left[-2x,\ -x,\ \frac{1}{6}\right]$	[12,15]

Table 11: Wyckoff site: 24k, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x-y,\ x,\ z+\frac{5}{6}\right]$	[2]
3	$\left[-y,\ x-y,\ z+\frac{2}{3}\right]$	[3]
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[4]
5	$\left[-x+y,\ -x,\ z+\frac{1}{3}\right]$	[5]
6	$\left[y,\ -x+y,\ z+\frac{1}{6}\right]$	[6]
7	$[x-y,\ -y,\ -z]$	[7]
8	$\left[y,\ x,\ \frac{2}{3}-z\right]$	[8]
9	$\left[-x,\ -x+y,\ \frac{1}{3}-z\right]$	[9]
10	$\left[x,\ x-y,\ \frac{5}{6}-z\right]$	[10]
11	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	[11]
12	$\left[-y,\ -x,\ \frac{1}{6}-z\right]$	[12]
13	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[13]
14	$\left[x-y,\ x,\ z+\frac{1}{3}\right]$	[14]
15	$\left[-y,\ x-y,\ z+\frac{1}{6}\right]$	[15]
16	$[-x,\ -y,\ z]$	[16]
17	$\left[-x+y,\ -x,\ z+\frac{5}{6}\right]$	[17]
18	$\left[y,\ -x+y,\ z+\frac{2}{3}\right]$	[18]
19	$\left[x-y,\ -y,\ \frac{1}{2}-z\right]$	[19]
20	$\left[y,\ x,\ \frac{1}{6}-z\right]$	[20]
21	$\left[-x,\ -x+y,\ \frac{5}{6}-z\right]$	[21]
22	$\left[x,\ x-y,\ \frac{1}{3}-z\right]$	[22]
23	$[-x+y,\ y,\ -z]$	[23]
24	$\left[-y,\ -x,\ \frac{2}{3}-z\right]$	[24]